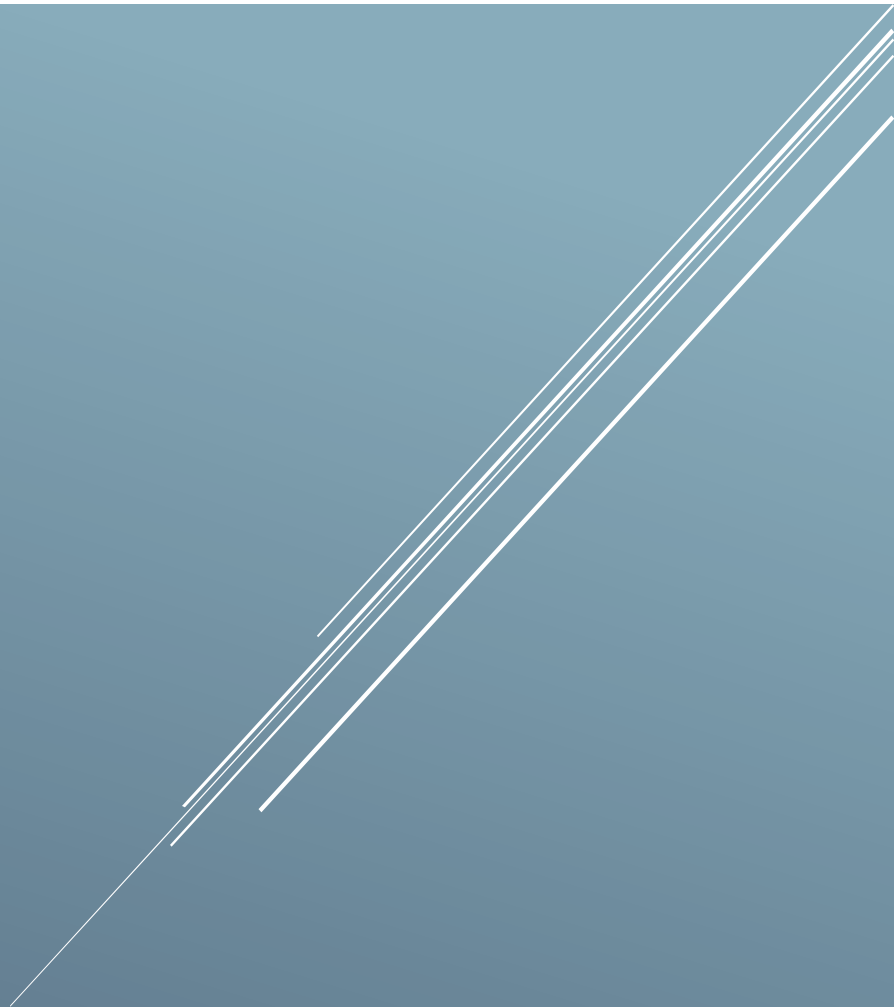




CUSTOMER CHURN PREDICTION

APPLIED DATA ANALYSIS ASSIGNMENT 2

GROUP 13

KAGGLE ID: GROUP 13

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1. Introduction

Customer churn is a pivotal phenomenon that reflects the rate at which customers discontinue their association with an organization. It holds significant importance for businesses across various sectors, as it directly impacts the financial health, sustainability, and profitability of companies. The central idea behind customer churn is to understand why customers, who were once engaged with a company's products or services, decide to terminate that association, and take appropriate action at the right time.

In this report, we are tasked with performing predictive analytics on a bank dataset to predict whether an existing customer will stop using the products of the bank or not. The findings are based on a classification task that predicts whether the customer churns or not based on the provided features. However, before moving on to the prediction task, the performed analysis of the data and the various predictors. The next section explains the data exploration and the feature engineering performed on the given dataset.

2. Exploratory Data Analysis

In our data exploration, we realized that our data is a mix of categorical, numerical, and textual data. As shown in Figure 1, most of the predictors are numerical with two nominal predictors 'has_credit_card' indicating whether the customer holds a credit card and 'is_active_member'. The 'customer_profile' represents text data, providing a detailed description of customers which we will later use to generate new potentially important predictors. Finally, the target binary variable 'Y' represents whether the customer churned or not.

Categorical Predictors		Numerical Predictors	Textual Predictor	Target Variable
<u>NOMINAL</u>	<u>ORDINAL</u>	credit_score tenure balance number_of_products salary	customer_profile	Y
has_credit_card is_active_member	No Ordinal Predictors			

Figure 1: Different types of predictors

The box plot shown in Figure 2 helps us to provide insights into the distribution of data within different features. When examining the set of features, **we can observe variations in the scales and ranges of values. Some features have small value ranges, while others span a wide range.** This discrepancy in scaling across features can impact the performance of machine learning algorithms. To address this issue, scaling methods such as Min-Max scaling or Standardization (Z-score scaling) can be employed. Min-max scaling adjusts the range of values to a common interval, typically between 0 and 1, while Standardization scales feature to have a mean of 0 and a standard deviation of 1. The scaling technique used in the final model is further explained in the later sections.

Additionally, it is evident from Figure 3 that the distribution of age of the customers who are not churning approximates a normal distribution with a mean of around 25. On the other hand, the distribution of customers who are churning has a mean of around 40. This suggests a **difference in behaviour between the two groups of customers.** Since we analyse the average age of customers, the class imbalance problem of one of the classes having a larger count of records will not affect this analysis. Further, we perform binning of the age column into 6 groups as shown in Figure 4. The stacked bar chart illustrates that the stable customers (Y = 0) mostly lie in the age range 30-40 while the leaving customers (Y = 1) lie in the age range 40-50. **This makes the predictor 'age' one of the significant predictors for our further analysis and predictive modelling.**

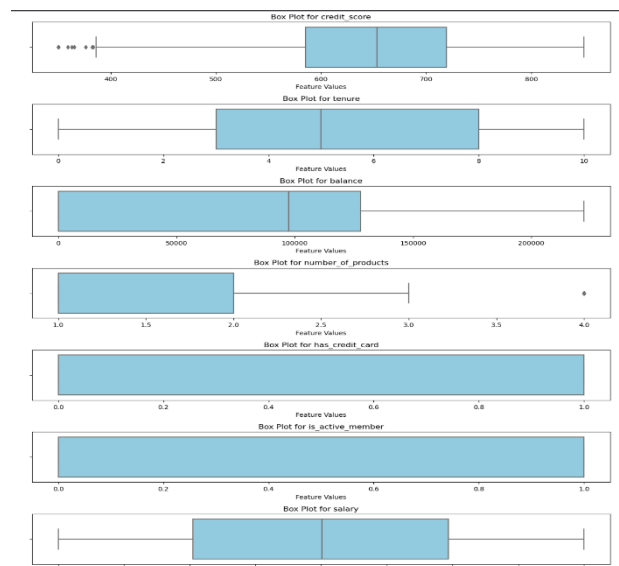


Figure 2: Box plot of features

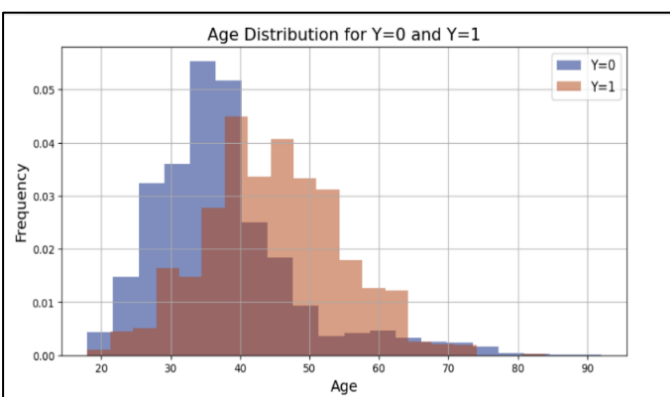


Figure 3: Age distribution for different target variables



Figure 4: Age distribution of Leaving and Stable Customers

3. Initial Feature Engineering

- The given column 'customer_profile' in the dataset consists of the textual description of the customer. It consists of the already given input variables. In addition to this, it consists of some new information about the customer such as age, gender, and country. To extract these features from the natural language text, we tried using regular expressions to extract

information from the text. Direct application of regular expressions did not work due to additional stopwords and different textual noises. Therefore, the following steps were taken to reduce the textual noise.

- Tokenization of the customer profile text into smaller pieces called tokens.
- Removal of stopwords from the tokenized text. E.g., of, an, the, etc.
- Lemmatizing the words to their base form. E.g., “products” converted to “product”, etc.

The above text preprocessing of the customer profile text also ensures that the regular expressions do not fail in case of unseen test data that the trained model might encounter in the future.

- The customer profile text is then ready for the application of regular expressions to fetch the age, gender, and country of the customers, thus creating three new features in the dataset. Additionally, we fetched the other customer information about the customer which is already provided in the dataset columns to validate them. **No inconsistencies were found between the given dataset and the information contained in the customer profile column.**
- Next, we removed some columns from the dataset such as ‘ID’ as it does not help in the prediction. Further, once the required data is extracted from the ‘customer_profile’ column, it is of no use in the text format and is removed from the dataset.
- Finally, the new columns ‘gender’ and ‘country’ which are nominal variables are converted into dummy variables. This created in total 3 dummy variables.
- For improvement of the classification model and the generator of more features, we used other techniques such as Feature scaling and Non-linear feature transformations which are explained in detail in the further sections.

4. Statistical Analysis using Logistic Regression

In our logistic regression analysis, several significant predictors were identified for customer churn using the following statistical analysis. For training the logistic regression model, we used all the predictors. The calculated coefficient estimates β_i , standard errors $SE(\beta_i)$, and the corresponding Z-statistic for each coefficient estimate β_i using the following equation are shown in Figure 5.

$$Z_i = \frac{\beta_i}{SE(\beta_i)}$$

Using the Z-values, and the corresponding p-values, we are trying to find the statistical evidence against the null hypothesis that $H_0: \beta_i = 0$. Utilizing low p-values as indicators of statistical significance, we found that credit score, tenure, balance, active membership status, age, and dummy variables for gender and country played crucial roles in predicting churn. These findings were derived from the coefficients associated with each variable. By incorporating these significant variables, our churn prediction model can be refined to be more accurate.

Logit Regression Results						
Dep. Variable:	Y	No. Observations:	5600			
Model:	Logit	DF Residuals:	5586			
Method:	NLE	DF Model:	13			
Date:	Sun, 22 Oct 2023	Pseudo R-squ.:	0.1515			
Time:	01:39:46	Log-Likelihood:	-2426.4			
converged:	True	LL-Null:	-2859.5			
Covariance Type:	nonrobust	LLR p-value:	8.915e-177			
	coef	std err	z	P> z	[0.025	0.975]
const	-2.6700	0.404	-6.607	0.000	-3.462	-1.878
credit_score	-0.0009	0.000	-2.299	0.022	-0.002	-0.000
tenure	-0.0277	0.012	-2.216	0.027	-0.052	-0.003
balance	2.46e-06	6.88e-07	3.576	0.000	1.11e-06	3.81e-06
number_of_products	-0.0854	0.064	-1.344	0.179	-0.210	0.039
has_credit_card	-0.1718	0.103	-1.672	0.095	-0.373	0.030
is_active_member	-1.0823	0.179	-6.030	0.000	-1.434	-0.731
salary	4.49e-07	6.32e-07	0.710	0.478	-7.9e-07	1.69e-06
age	0.0708	0.003	21.067	0.000	0.064	0.077
customer_profile_polarity	-0.7584	0.394	-1.924	0.054	-1.531	0.014
last_lines_polarity	0.4493	0.243	1.851	0.064	-0.026	0.925
gender_male	-0.5200	0.072	-7.191	0.000	-0.662	-0.378
country_Germany	0.9028	0.090	10.046	0.000	0.726	1.078
country_Spain	0.0987	0.095	1.042	0.297	-0.087	0.284

Significant Coefficients:	
const	-2.670033
credit_score	-0.000852
tenure	-0.027661
balance	0.000002
is_active_member	-1.082348
age	0.070763
gender_male	-0.519968
country_Germany	0.902026
dtype:	Float64

Figure 5: Statistical Analysis using Logistic Regression

5. Class Imbalance problem and its solution

Class imbalance in labeled datasets is a common challenge in machine learning, where one class has significantly fewer instances compared to another. This imbalance can lead to biased predictions, especially when the minority class is of interest. To explore the **negative impact of the class imbalance problem**, we trained a logistic regression model that predominantly predicted the majority class. We explore the impact of this imbalance on the model's performance and explain how different sampling methods can be applied to mitigate this issue. In the given dataset, the target column Y has two classes, class 0 with 5565 instances and class 1 with only 1435 instances. When we trained a logistic regression model on this data, **the results were skewed** and the model overwhelmingly predicted the majority class, resulting in a confusion matrix as shown in Figure 6. It shows a true negative metric of 1663 and a false negative metric of 437 meaning that the results were biased towards the negative class. Clearly, the model was not effectively capturing the minority class.

[[1663	0]
[437	0]]

Figure 6: Confusion Matrix without SMOTE

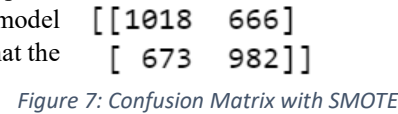
Three common techniques can be used to handle class imbalance problems: over-sampling, under-sampling, and Synthetic Minority Over-sampling Technique (SMOTE).

Over-sampling: This technique increases the number of instances in the minority class by duplicating or generating new samples. The goal is to balance the number of samples between the minority class (label 1) and the majority class. It does so by generating additional instances of the minority class through a process known as bootstrapping. Bootstrapping involves randomly sampling with replacements from the existing minority class samples to create new instances.

Under-sampling: It aims to balance the distribution by reducing the number of instances in the majority class. It achieves this by randomly removing samples from the majority class. By doing so, it aims to make the model focus more on the minority class. The

process of under-sampling helps prevent the model from being biased toward the majority class and can lead to more balanced and unbiased predictions, particularly for the minority class.

SMOTE: SMOTE (Synthetic Minority Over-sampling Technique) is a specialized technique used to balance the class distribution in machine learning. SMOTE doesn't merely duplicate or remove instances, but instead, it creates synthetic data points for the minority class. This is achieved by interpolating between adjacent samples in the feature space. SMOTE generates new data that reflects the underlying patterns of the minority class, providing the model with a more balanced dataset. This technique helps to reduce the risk of overfitting while ensuring that the model pays greater attention to the minority class, ultimately leading to more accurate and unbiased predictions for both classes.



As shown in Figure 7, the confusion matrix is improved by the application of the SMOTE technique. However, the False positives and False Negatives are still high because we have used the basic logistic regression model to fit the non-linear data.

Out of these explored techniques, the SMOTE worked best for the given dataset.

6. Project Management

In order to collaborate with our team members during this group project, we used the **Trello Board** which allowed for task simplification (dividing a broad process into specific and easily manageable tasks) and thereby, facilitated effective project management. Additionally, it provided us with a space where we shared and monitored the team's progress and challenges. During the period of four weeks,

- We consistently used Trello for defining our tasks and tracking the progress of the project.
- Scheduled weekly sprint meetings as well as daily standup meetings which proved to be a highly efficient process in the completion of this project.

The following table shows the summary as well as the hours distribution for the different tasks defined during each week.

Member	Week	Task	Trello Link	Salary	Member	Week	Task	Trello Link	Salary
Abhishek	9	Project Management	-	\$250	Lokesh	11	Project Management	-	\$250
Abhishek	9	Planning the project, Reading and understanding the Project at hand, Trello tutorial	https://trello.com/c/nK8q96ww	\$1000	Lokesh	11	Create a combined boilerplate notebook	https://trello.com/c/ff4yKyRv	\$200
Rohin	9		https://trello.com/c/yzslvRU3	\$1000	Rohin	11	Statistical Analysis of logistic regression results	https://trello.com/c/yB1HRpPJ	\$500
Lokesh	9		https://trello.com/c/vi08KmvJ	\$1000	Rohin	11	Logistic Regression implementation	https://trello.com/c/hRhN9DBp	\$500
Rohan	9		https://trello.com/c/MSFO9vmR	\$1000	Lokesh	11	Application of DNNs and RNNs	https://trello.com/c/aec57XRq	\$200
Rohin	10	Project Management	-	\$250	Rohin	11	Feature Selection and scaling	https://trello.com/c/s6UQUWyt	\$500
Abhishek	10	Extracting features from Customer profile	https://trello.com/c/ulyOOFPq	\$300	Lokesh	11	Random forest implementation	https://trello.com/c/jlt1l0Cn	\$300
Abhishek	10	Regex on given Customer Profile	https://trello.com/c/slwe5WVr	\$200	Lokesh	11	Gradient Boosting Algorithm	https://trello.com/c/DwYB0gkQ	\$300
Abhishek	10	Regex on pre-processed customer profile	https://trello.com/c/UUNzioUc	\$500	Lokesh	11	SVM implementation	https://trello.com/c/yJh7YlP	\$500
Lokesh	10	NLP on Customer Profiles Column	https://trello.com/c/c7HSebh	\$100	Lokesh	11	Using SMOTE for training classifiers	https://trello.com/c/PS2Hzt0L	\$500
Lokesh	10	Sentiment Analysis on Customer Profiles	https://trello.com/c/uToAnG4P	\$200	Rohan	12	Project Management	-	\$250
Abhishek	10	EDA: Correlation of different columns with the target	https://trello.com/c/8baVzPDD	\$500	Abhishek	12	Variable importance for model improvement	https://trello.com/c/trDuxra3	\$1000
Lokesh	10	EDA: Sentiment Analysis features	https://trello.com/c/Zar6mhKo	\$200	Rohan	12	Feature transformations for models' improvement	https://trello.com/c/trDuxra3	\$500
Rohin	10	Outlier detection and removal	https://trello.com/c/hfR7J0Qu	\$500	Rohin	12	Hyperparameter tuning and models' improvement	https://trello.com/c/trDuxra3	\$200
Rohin	10	Missing data Analysis	https://trello.com/c/AaN4bk82	\$500	Abhishek	12	Report part A	https://trello.com/c/kky8xC2f	\$500
Rohan	10	Research on sampling methods	https://trello.com/c/v8MFQRT	\$750	Lokesh	12	Report part B	https://trello.com/c/nPVxXFat	\$500
Rohan	10	Implementation of Sampling Methods	https://trello.com/c/ohFaMfph	\$750	Rohan	12	Report part C	https://trello.com/c/ZyWCEfCK	\$500
Rohan	10	Common function for sampling methods & hyperparameter tune	https://trello.com/c/a0aHRwES	\$500	Rohin	12	Report part D	https://trello.com/c/YLd4jua5	\$300
								https://trello.com/c/rz1ZOK9G	

7. Model Development and Selection

In order to predict customer churn accurately, we explored three different machine learning models, each with its own unique characteristics and assumptions. Here's a brief overview of the models we tested, along with their respective performances.

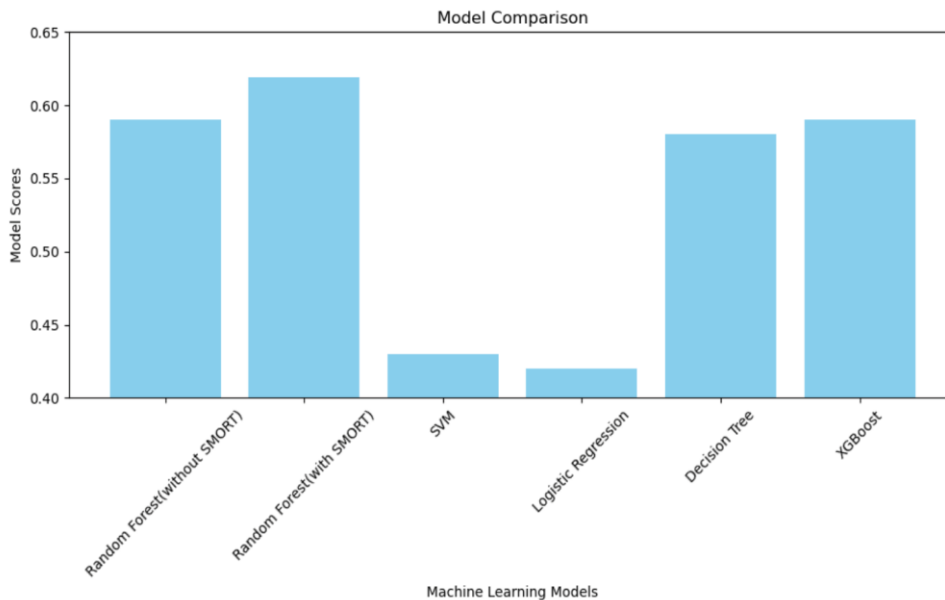


Figure 8: Model f1 score comparison

Logistic Regression: Logistic regression, a fundamental classification algorithm, assumes a linear relationship between input features and the binary churn target variable. Despite its simplicity, logistic regression yielded less satisfactory results, with an F1 score of 0.42. This suggests that the straightforward linear approach may not have been adequate for capturing the complexity of our churn prediction task.

Support Vector Machine: SVM, a powerful machine learning algorithm, works by finding the best boundary that separates different classes in the data. It's known for its versatility and ability to handle complex data. In our experiments, SVM demonstrated its strength, achieving an F1 score of 0.43. While not the top performer, it showed its competence in addressing our churn prediction task, making it a valuable addition to our model selection.

Since, the data is highly non-linear as explained in the above sections, we moved to some non-linear models for fitting the dataset

Decision Tree Algorithm: Decision Tree is a fundamental machine learning algorithm that recursively divides data into subsets based on the most significant attribute. It is a simple yet powerful model for both classification and regression tasks. In our experiments, Decision Trees offered an interpretable insight into feature importance and decision-making processes. While not as complex as Random Forest, Decision Trees can still capture essential patterns in the data. In our churn prediction task, Decision Trees delivered an F1 score of 0.58, demonstrating their competence in handling the problem. rephrase in easy language.

Random Forest: Random Forest, an ensemble learning method, constructs numerous decision trees and averages their predictions. It excels in capturing intricate patterns and avoiding overfitting. In our experiments, Random Forest stood out as the top performer, achieving an F1 score of 0.62. Its ability to handle complex relationships made it the ideal choice for our churn prediction task.

XGBoost (Extreme Gradient Boosting): XGBoost, a gradient boosting algorithm, builds a sequence of decision trees to optimize predictions. While highly efficient, it trailed slightly behind Random Forest in our scenario, achieving an F1 score of 0.59. Despite its strong predictive accuracy, it did not outperform Random Forest in our churn prediction task.

To summarize, our analysis led us to select the Random Forest model due to its robust performance in capturing intricate patterns and avoiding overfitting. It outperformed Logistic Regression, SVM, Decision Tree and XGBoost, making it the best solution for our churn prediction task.

Model Development Steps:

1. SMOTE Sampling: To address the class imbalance issue in our dataset, we employed the Synthetic Minority Over-sampling Technique (SMOTE). This technique helps balance the class distribution by generating synthetic samples for the minority class, ensuring a more representative dataset. SMOTE goes beyond simple duplication or removal of instances; instead, it generates synthetic data points for the minority class by interpolating between neighboring samples in the feature space. This process creates new data that captures the inherent patterns of the minority class, resulting in a more balanced dataset. By using SMOTE, you reduce the likelihood of overfitting and ensure that the model focuses on the minority class, leading to more precise and unbiased predictions for both class categories.

2. Train-Test Split: We divided the preprocessed dataset into training and testing sets, allocating 80% of the data for training and 20% for testing. This separation allowed us to train the model on one subset and evaluate its performance on unseen data, ensuring an unbiased assessment.

3. Hyperparameter Tuning: Utilizing GridSearchCV, we systematically explored the hyperparameter space of our chosen classification algorithm (such as Random Forest or XGBoost). This randomized search approach efficiently sampled hyperparameters, enabling us to identify the optimal configuration for our model. Hyperparameter tuning was crucial for enhancing the model's predictive power and generalizability. Following are the tuned hyperparameters of the random forest classifier (one of the best classifier models).

When creating an ensemble of multiple decision trees with a specified number of estimators (`n_estimators`), the algorithm doesn't consider all predictor features at each split. Instead, it selects only a subset of predictors, and this subset size is determined by the `max_features` hyperparameter. This approach is employed to reduce the correlation between the trees in the ensemble. To optimize the performance of the ensemble, the hyperparameters **`n_estimators`** and **`max_features`** are fine-tuned using a Grid Search Cross-Validation approach. This involves trying different combinations of values for these hyperparameters, specifically:

For `n_estimators`, values {100, 500, 1000} are tested.

For `max_features`, values {10, 20, 50, 100, 200, 'ALL'} are explored.

This systematic tuning process helps find the optimal configuration for the ensemble of decision trees, enhancing its predictive accuracy and robustness with `n_estimators` = 1000 and `max_features` = 10.

4. Pipeline Construction:

- **Standard Scaling:** The features in the dataset were standardized using the StandardScaler. Standardization ensures that all features have a mean of 0 and a standard deviation of 1, which is crucial for models sensitive to the scale of input features.
- **Polynomial Features:** To capture potential non-linear relationships, we applied PolynomialFeatures. This transformation generated polynomial and interaction features up to the degree of 2, allowing the model to account for complex patterns in the data.
- **Classifier Integration:** We created a pipeline that included the StandardScaler, PolynomialFeatures, and the chosen classification algorithm (such as Random Forest or XGBoost). This integrated approach streamlined the process, ensuring consistent preprocessing steps for both training and testing data.

5. Model Training and Evaluation: The pipeline, comprising the preprocessing steps and the classifier, was trained on the training dataset. To assess the model's performance, we utilized the **F1 score**, a metric that balances precision and recall. This metric provided a comprehensive evaluation of the model's ability to correctly predict both positive and negative classes, offering a more nuanced understanding of its effectiveness.

8. Conclusion and Discussion

This project aimed to predict customer churn in the banking sector, a critical factor affecting a company's financial health. Leveraging predictive analytics, our primary goal was to determine if existing bank customers would discontinue product usage, utilizing various features.

Our journey began with comprehensive data exploration, revealing a mix of categorical, numerical, and textual data. Extensive exploratory data analysis (EDA) provided insights into feature distributions and scale disparities, leading to the application of techniques like Min-Max scaling and Standardization to standardize features.

Crucial predictors emerged, including customer age, and age-based binning offered nuanced insights into different age group behaviors. We also conducted initial feature engineering by extracting valuable information from the 'customer_profile' text column, enriching our dataset.

Statistical analysis using logistic regression identified vital predictors that improved our churn prediction model. Addressing class imbalance, we employed the Synthetic Minority Over-sampling Technique (SMOTE) to balance class distribution.

In the model development phase, we evaluated various models, ultimately selecting Random Forest due to its superior F1 score of 0.62. Random Forest consistently outperformed other models in our experiments, making it the ideal choice for our churn prediction task.

In conclusion, our selection of the Random Forest model was informed by extensive exploratory data analysis and rigorous model development, leading to superior performance in predicting customer churn accurately. This choice reflects our commitment to selecting the most robust model for this crucial task.

Appendix:

Trello Cards:

Group 13 Workspace visible Board Board

Filters RG AS LA RS Share

Backlog

To do in Week 12

Doing in Week 12

Done in Week 12

- Logistic regression and statistical analysis in report
3 RG
- Project management part in report
LA
- Improving the EDA part in report
1 AS LA RS
- Writing model development and selection part in report
2 RG
- Model Improvement
3 LA AS RS RG
- Writing the introduction as well as EDA part in the report.
1 AS

Class imbalance problem in report

☰ 1

RS

Data preprocessing and initial feature engineering in report

☰ 1

LA

AS

Age Graph to be added in EDA

☰ 1

AS

To do in Week 11

Doing in Week 11

Done in Week 11

Create a combined boilerplate notebook

Logistic Regression implementation

Feature Selection and scaling

Application of Deep neural networks and Recurrent Neural networks

Basic Random forest implementation

Using SMOTE for training classifiers

Applying Gradient Boosting Algorithm

Basic Decision tree implementation

Basic SVM implementation

Statistical Analysis of logistic regression results

EDA for different features

To do in Week 10

Doing in Week 10

Done in Week 10

Extracting features from Customer profiles

EDA: Correlation of different columns with target variable

EDA: Sentiment Analysis

Common function for sampling methods and hyperparameter tuning

Regular expression on pre-processed customer profile column

Regular Expression on Customer Profile Column

Implementation of Sampling Methods

Research on sampling methods

Missing data Analysis

Sentiment Analysis on customer profiles

Natural Language Processing on Customer Profiles Column

Outlier detection and removal

To do in Week 9

Doing in Week 9

Done in Week 9

Trello tutorial

1



AS

LA

RS

RG

Reading and Understanding the
Project at hand

1



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LA

RS

RG

Project planning

2



RG

AS

LA

RS

Assigning tasks

/trello.com/b/rdp7B0tl/group-13

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Group 13 | Trello

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